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Looking for complementarities. Within-industry diversification and geographic diversification of Venture Capital Firms

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ABSTRACT

Research on Venture Capitalists' (VCs) industrial diversification is supplemented with the notion of diversification along an industry's value chain. VCs are hypothesized to create a portfolio of complementary investments along the value chain, accompanied by low geographic diversification. Further, VCs specializing in an industry with network externalities are predicted to devote relatively more investments to this industry, followed by an increased propensity for diversification along the value chain. This, subsequently, will result in less geographic diversification. The hypotheses are supported through a study on VCs in the US and Europe. Contributions to the literature on VCs, diversification, geographic agglomeration and network effects are discussed.

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Introduction

Why do Venture Capitalists diversify within industries, when risk might be spread more efficiently across industries, and why do they concentrate their investments locally, rather than maximizing their resource pool through dispersing their investments around the globe? The goal of this paper is to inquire into this question by, first, supplementing the existing typology of industrial diversification through diversification along the value chain, and, second, by theorizing the relationships between industrial and geographic diversification. So far, diversification of Venture Capital firms (VCs) has been studied both from a financial investment lens (Cressy, Malipiero, and Munari 2014; Knill 2009) and from a strategic management perspective (Matusik and Fitza 2012; Yang, Narayanan, and De Carolis 2014). Generally, there is much knowledge of VCs risk-return and growth considerations and on their choice of types of diversification, mostly based on distinctions along the continuum between related and unrelated diversification (Rumelt 1982). Little attention so far found VCs' diversification along an industry's value chain. As I will show in this paper, conditions for realizing superior returns with industrial diversification differ across industries and are dependent on the degree of geographic diversification, thus calling for an integration of these largely independent streams of research.

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Extant research has already supplemented the related-unrelated continuum by more nuanced forms of diversification along stages of firm development (early to late stage) (Zhang 2014). In addition, studies examined effects of degrees of syndication with other VCs (Brander, Amit, and Antweiler 2002; Sorenson et al. 2001). Further, VCs usually subscribe either to a passive or active style of investing (Macmillan, Kulow, and Khoylian 1989), which appears to moderate the relationship between diversification strategy and VCs performance (Bottazzi, Da Rin, and Hellmann 2008). While venture capital investing is highly concentrated to certain types of industries, such as software and biotechnology (Tarhuni and Black 2016), neither theory nor empirical work on VCs' diversification so far account for the type of industry, which will be done in this paper by distinguishing industries in regard to the presence of network externalities.

Virtually unrelated to this stream of research are findings that VC investment is all but randomly distributed geographically (see for example Figure 1). This is especially attributed to the fact that VCs invest to a large proportion into specialized regions or clusters such as the Silicon Valley (Engel 2015) or the US East Coast, with the latter dominated by firms engaged in the Life Sciences (Kolympiris and Kalaitzandonakes 2013).

These cases already suggest a link between VCs' diversification strategy, industry conditions and geographic concentration. So far, however, there is neither theoretical nor empirical work to explain benefits due the systematic combination of industry characteristics, the specific diversification strategy of Venture Capitalists and their associated geographic diversification. Thus, the intended contribution of this paper is to provide a theoretical explanation for the mutual benefits of within-industry diversification and geographic concentration, which is based on the notion of complementarity (Milgrom and Roberts 1995). Of special relevance in the context of this paper is the notion of the *value chain*, which is characterized through complementary activities. The findings of the paper support the derived hypotheses that, first, more VCs' investments into an industry is followed by increased diversification along the value chain and, second, that this is accompanied through reduced geographic diversification of investments. Further, I will show that the benefits of complementary investments can be enjoyed especially in industries which are characterized through network externalities. This gives rise to the third contribution to explain differences across industries in regard to combinations of industrial diversification and geographic diversification. The results of the present study indeed show that the above effects are stronger in digital industries compared to the life and health sciences, which are used here to represent differences in regard to the existence of network externalities.

The paper proceeds as follows. The first section reviews extant empirical research in regard to VCs' industrial diversification and geographic diversification. The second section provides the theoretical basis by using the notion of complementarity, by proposing that complementarities are especially salient under the presence of network externalities, and that all of this calls for geographic concentration of investments. The third section contains the empirical test of the thus derived hypotheses, followed by a final discussion.

(a) Europe and Asia



(b) Regional Clusters in the US

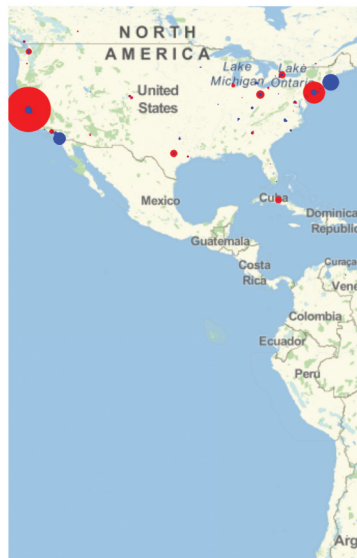


Figure 1. Regional clusters of VC investments. (Size of dots indicate degree of agglomeration, Red for Digital Industries, Blue for Life Sciences). (a) Europe and Asia. (b) Regional Clusters in the US.

Research on diversification of Venture Capital Firms

Industrial diversification

Similar to corporations, VCs have to trade-off opposing forces with their diversification strategies: The distribution of risk, calling for unrelated diversification, and the realization of economics of scope, for example through the acquisition of certain types of knowledge by VCs, as a main driver for related diversification, especially for within-industry diversification (Cressy, Malipiero, and Munari 2014; Hashai 2015; Matusik and Fitza 2012; Neffke and Henning 2013). According to results obtained by Yang, Narayanan, and De Carolis

(2014), the mere degree of diversification (based on the diversity of SIC-Codes, without distinguishing relatedness) for corporate venture capital relates in an U-shaped manner with performance. In contrast, looking on failure rates of VC's portfolio firms, Makarevich (2018) finds an inverted U-shaped relationship with specialization, especially with a higher failing propensity for generalist VCs.

Similarly, several arguments suggest limits to VCs unrelated diversification. First, industry specific knowledge helps to reduce uncertainty, which is especially high when deciding upon investing into new firms (Matusik and Fitza 2012). Over time, when repeatedly investing in an industry, VCs develop information networks, social ties, and alliances to gain superior judgment about investment opportunities (Sorenson et al. 2001). Second, to the extent that VCs actively interfere in their portfolio companies, experience from similar companies can be shared to develop new enterprises (Bottazzi, Da Rin, and Hellmann 2008). Third, facilitated by low geographical distance between the venture capitalist and the investee firms (Lutz et al. 2013), active investors may utilize industry specific expertise for controlling, monitoring and reducing agency problems (Bellavitis, Kamuriwo, and Hommel 2019; Sahlman 1990). Starting with an early questionnaire study (Macmillan, Kulow, and Khoylian 1989), however, systematic differences in regard to the level and type of VCs' involvement in their investee firms could only be found along the life cycle of VCs' investment (e.g., Berglund 2011). Fourth, resources beyond knowledge can be shared with less coordination and adaptation costs between related companies compared to companies from different industries (Lerner 1995), at least for firms with organizational slack (Gary 2005). Finally, intra-industry diversification likely leads to multiple point competition (Karnani and Wernerfelt 1985) or multi market contact among rival firms, which has been argued to increase possibilities for collusion and mutual forbearance, thereby positively influencing performance.

Geographic diversification

Generally, returns to diversification are exploited independent of the geographic location of enterprises. Accordingly, there are few studies which examine the link between industrial diversification and geographic diversification. Exemptions are early studies of diversified companies which found positive effects of the degree of divisionalization (which is a consequence of diversification) on geographic dispersion (Grinyer, Yasai-Ardekani, and Al-Bazzaz 1980). Similarly, to maximize their resource pool VCs should diversify geographically. But the majority of VCs rather concentrate their investments to a limited range of regions (Subhash 2007), especially when specializing in early-stage start-ups and small ventures (Gupta and Sapienza 1992). As an explanation, geographic concentration of human capital and of competencies has been proposed already by Marshall (1898). For the case of VCs, the fact that a large proportion of Venture Capital flows into California is largely ascribed to concentration of skilled labour and clustering effects (Engel 2015). Country differences in regard to the availability of appropriate institutional regulations have also been identified as a limitation (Bellavitis, Kamuriwo, and Hommel 2019), whereas Moore et al. (2015) found regulatory differences to be less restricting to cross-border diversification than normative and cultural factors. In addition to such supply side effects (especially human resources) demand side benefits (for example by easing customer search) also may favor regional agglomeration (McCann

and Folta 2008). Related additional explanations for geographical proximity as a driver of innovativeness and performance within regional clusters (Audretsch and Feldman 1996) refer to knowledge spillovers (Plummer and Acs 2014), increased motivation through competition, and direct interaction between firms and their members (Feldman 1994; Molina-Morales, Garcia-Villaverde, and Parra-Requena 2014; Taylor and Spicer 2007), thus potentially mitigating the risks of high-growth new entrants (Pe'Er, Vertinsky, and Keil 2016). Especially under high ambiguity firms rely on social cues of others in their neighbourhood, thus encouraging new venture creation where others have been started already (Chang, Chrisman, and Kellermanns 2011; Minniti 2005).

Especially with the latter, geographical proximity is interpreted as a way to overcome market limitations by introducing organizational principles of reducing uncertainty and ambiguity (Daft and Weick 1984) into seemingly independent firms. First, if VCs hold portfolio firms from the same geographic region, those may more rely on personnel with similar cultural and educational background. Employees frequently switch between companies within the same industry and geographic area. Second, despite the existence of electronic messaging systems and video conferencing infrastructure, face-to-face meetings to reduce ambiguity and uncertainty (Daft and Lengel 1986), workshops, and conferences to exchange industry experiences remain important, which also will be more likely and frequent when travel time is short. Such can be exploited in markets by firms forming communities (Wade 1995), with a cohesive network structure (Funk 2014), thus enhancing the likelihood for cooperation (Powell et al. 2005) and collusion (Gan and Hernandez 2012), but also of unintended knowledge spillovers to rivals (Plummer and Acs 2014; Ryu, McCann, and Reuer 2018). Geographic proximity allows for more face-to-face interaction to reduce ambiguities, which are especially salient when new products shall be launched, new markets need to be created for a value chain, and when innovating in firm networks (Schilling and Phelps 2007). Also, new firms find more easily investors if similar firms operate in their neighbourhood (Kolympiris, Kalaitzandonakes, and Miller 2011), while diversification by country showed positive effects on VC performance for UK-based VCs (Cressy, Malipiero, and Munari 2014). In the following theoretical section, such somewhat conflicting findings on geographic diversification shall be partly resolved and explained through its relation to industrial diversification and through the degree to which network externalities exist in industries.

Theory: Reasons for Venture Capital diversification

Types of diversification

Theorizing and empirical research on diversification is essentially guided, first, by defining degrees of diversification and, second, by distinguishing types of diversification, which represent different solutions either for spreading risk or for trading off economies of scale and scope. As the limits to unrelated diversification are well researched by now, the focus has been turned more recently to within-industry diversification (Barbiroli and Focacci 2005; Hashai 2015; Li and Greenwood 2004). In an even more fine grained manner, Tanriverdi and Chi-Hyon (2008) distinguished complementary investments within an industry (in their case software platform investments) from pure market related diversification (different software applications for the same customer base). Based on Milgrom

and Roberts (1995) concept of complementarity, they argue that cost synergies through sharing common resources can be exploited only when diversifying in a related manner across applications on common platforms. Accordingly, they found positive effects on returns in the software industry for firms combining complementarity with market related diversification.

The concept of complementarity provides also the appropriate basis for this paper to develop the following two theoretical arguments as a basis for subsequent hypotheses: First, (dis-)economies of scope for VCs have to be distinguished from those of diversifying corporations. Traditionally, economies of scope are thought of reducing costs through the use of resources for a diverse set of products or services, for example by combining resources for new partnerships or through relational contracting (Lindsey 2008). Much more important for VCs, however, is the potential increased output due to economies of scope, as a result of more or less free added value by utilizing scale-free resources, such as knowledge (Sakhartov 2017). Second, economies of scope differ significantly between two types of within-industry diversification: (a) Pure related diversification within an industry, to be labelled as within-segment diversification, and (b) diversification along the value chain. In the following I explain shortly the concept of complementarity, to be used as a basis for this distinction and as a basis for the subsequent formulation of hypotheses in regard to diversification along the value chain.

Complementarity as a source of value

Activities are complementary “if doing (more of) any one of them increases the returns to doing (more of) the others” (Milgrom and Roberts 1995, 181), resulting in a supermodular return function, which is a mathematical formulation for synergies: If products (features) a_0 and b_0 are improved to a_1 and b_1 , then the sum of these improvements ($f[a_1, b_1] - f[a_0, b_0]$) is more than the sum of individual improvements ($f[a_1] + f[b_1] - f[a_0, b_0]$). This theory is formulated for ordered sets (lattices), leading to the notion of a *chain* (ibid, p.181), surfacing in markets as *value chains*. Therefore, the concept of the value chain and diversification along the value chain provides the core for this paper. Mathematically, complementarity requires that each feature (product, firm etc.) must be represented by a sub-lattice of the total lattice. Although by far not restricted to, industrial value chains by definition are characterized by supermodular return functions: for example (a) producing parts of a vehicle *and* (b) assembling it to a vehicle, together has more value than the sum of parts and assemblage ($f[a, b] > f[a] + f[b]$). In contrast, for producing different variants for the same part of a vehicle (no (re-)ordering possible for which the above conditions of lattices hold) usually no such supermodular function exists. Supermodularity even preserves its positive joint impact of individual improvements under uncertainty, as long as environmental shocks (such as technological or economic changes) affect all parts of the chain in the same way (Milgrom and Roberts 1995, 186).

Supermodularity is enhanced through further features of value chains. While producers of parts usually aim at consumers as customers, the assembling firm target business firms. Such a distinction holds generally for different stages (the ordered set) along the value chain. Upstream firms in the value chain may act as push factors for downstream developments. Conversely, growth in downstream firms pulls upstream firms, thereby mutually increasing revenues and subsequent returns for investors. In contrast to pure market

related diversification, which aims at a singular customer base, complementary investments located along a value chain, from upstream products (basic parts, platforms) to downstream products or services (consumer products or investment products, final services), serve distinct groups of customers, from businesses to consumers. With more volume, upstream products also gain in legitimacy, subsequently contributing to value and demand for downstream products which are built on them. Although such value chains may range across a diverse set of industries, major parts are contained within traditionally defined industries (see Table 1 for two examples, which will be used in the empirical section of this paper). For example, the value chain in the IT-Industry ranges from manufacturing of computers (NACE 26), over wholesale (465), retail (474), to programming (62) and information service activities (63).

While complementarity along the value chain exists independent of cooperation, it provides an incentive for it. While this is little researched for VCs, with one exception for the VC-investee relationship (Maula, Autio, and Murray 2009), studies in the biotechnology industry (Baum, Calabrese, and Silverman 2000; Baum and Silverman 2004; Rothaermel and Deeds 2004) found frequent cooperation of young innovative firms with established firms along the value chain (upstream and downstream alliances) to further develop products. This has been characterized as exploration alliances vis-à-vis exploitation alliances (Koza and Lewin 1998).

These positive effects of complementary investments along the value chain are contrasted by the finding of vertically integrated firms or corporations, which cover the whole range of value chains, to perform comparatively inefficient. Such is attributed to diseconomies of scope – firms not being able to efficiently handle the complexity of its parts (Palich, Cardinal, and Miller 2000), which might call for increased centralization of decision-making (Brahm and Tarziján 2015). VCs diversifying along the value chain, in a sense, are mimicking the structure of both horizontally and vertically diversified corporations, which tend to outsource downstream operations to reduce diseconomies of scope (Rawley and Simcoe 2010). This is a form of vertical quasi-integration which has been observed long ago (Blois 1971; Eccles 1981). Even more so and although

Table 1. Definition of value chains (based on Invest Europe, 2007).

NACE	Definition
(a) Digital Industry	
26	Manufacture of computer, electronic and optical products
465	Wholesale of information and communication equipment
474	Retail sale of information and communication equipment in
62	Computer programming, consultancy and related activities
63	Information service activities
9511	Repair of computers and peripheral equipment
(b) Health/Life Sciences Industry	
266	Manufacture of irradiation, electromedical and electrotherapeutic equipment
21	Manufacture of basic pharmaceutical products and pharmaceutical preparations
325	Manufacture of medical and dental instruments and supplies
4774	Retail sale of medical and orthopaedic goods in specialized stores
721	Research and experimental development on natural sciences and engineering
86	Human health activities
87	Residential care activities
88	Social work activities without accomodation

loosely coupled, portfolio firms together with its VC may act like quasi-firms (Luke, Begun, and Pointer 1989), with a common strategic purpose (e.g., setting standards, developing a field), more or less explicitly defined by the VC. In contrast to vertically integrated corporations, however, VCs are able to exploit benefits from cooperation among their portfolio firms, without carrying the costs of cooperation, which usually accrue for vertically integrated corporations. The reason lies in the “business model” of VCs. While VCs effectiveness in selecting firms may be diminished by an attribution error resulting in an overemphasis of firms’ human capital, they appear effective in assessing the potentials of alliances among these firms, in addition to their “coaching” in this regard (Baum and Silverman 2004). Even as portfolio firms may suffer from coordination costs with other firms, established valuation methods for venture capital systematically discount cost figures and rather focus on future value of the firm (Scherlis and Sahlman 1987), prospective earnings, customer counts, reputation (Hassan and Leece 2007), and judgments of management competence (Hassan and Leece 2007; Miloud, Aspelund, and Cabrol 2012). Still, as for vertically integrated firms, coordination costs likely lower performance in traditional accounting terms and even may lead to bankruptcy of some firms, but for the remaining firms VCs enjoy premium returns upon exit, because of such valuation methods. VCs searching for high returns, prefer high risk firms (Baum and Silverman 2004, 432) with little concerns for cost figures within these firms. As a result, diseconomies of scope are significantly lower for VCs than for vertically integrated firms.

Having discussed the potentials of complementarity for VCs, the second part of the argument rests on a distinction of within-industry diversification, which allows for the identification of conditions for harnessing these complementarity effects. First, within-segment diversification, (pure product-market related diversification) will result in a portfolio of firms competing for the same customer base and will imply a non-supermodular return function for combinations of products or services from these firms. Although VCs still may try to exploit benefits of related diversification through knowledge exchange, they likely do this at the cost of mutually threatening their portfolio firms. VCs will diversify within a market segment only if competition is mitigated by other strategies (price/quality discrimination) or by exogenous segmentation, especially through geographic separation. In contrast, as already defined above, products or services along the value chain target different groups of customers and constitute supermodular return functions. Therefore, VCs will prefer the second type of within-industry diversification, investing along the value chain, because such complementary investments do not necessarily result in competition among portfolio firms. Rather cooperation and alliances among portfolio firms will be encouraged. But in contrast to corporations and without actively interfering in portfolio firms, VCs are less likely to exploit traditional economies of scope by using common resources and reducing costs. Even without cooperation among portfolio firms, the notion of complementarity of investments, predicting supermodular returns, provides an incentive for VCs to invest into firms of the same industry, given that these firms are located on different stages along the value chain. Therefore, if VCs focus their investments to an industry, they will diversify more likely along the value chain than in a traditional within-segment manner. This leads to the following hypothesized relationship:

Hypothesis 1: The more portfolio firms VCs devote to an industry the higher the degree of within industry diversification along the single value chain of this industry (as opposed to within-segment diversification).

For this hypothesis we implicitly assumed that all industries contain value chains which allow to exploit complementarity to the same degree. In the next section this assumption is challenged by distinguishing industries in regard to the strength of potential complementarity.

Network externalities

The software industry so far provided one of the few empirical tests of the effects of complementary products and supermodularity (Lee, Venkatraman, and Tanriverdi 2010). Software serves as a prototypical example for industries characterized through network externalities since its early days (Gandal 1995). Products along the value chain are always to some extent complementary, but especially so under the presence of network externalities, as will be discussed in the following.

The literature distinguishes two types of network effects. First, indirect effects are due to the mutual dependence of products or services within an industry. Second, wherever the success of a product is dependent on the establishment of standards, or where users are interacting with products, the value of a product and of associated marginal returns increase with the number of customers – a relationship which is labelled as direct network effect (Birke 2009; Schilling 2002). In addition, the mutual dependence of products along the value chain should be enhanced especially through indirect network effects. For example, firms producing software applications are dependent on (upstream) companies which develop operating software or hardware platforms; and vice versa: Hardware and software platforms gain value only to the extent that consumer applications are based on them. As an example, the emerging sub-industry of Digital Currencies illustrates such dependencies: firms producing developer tools for firms programming business applications and firms establishing platforms for exchange (e.g., bitcoin exchange) act as hubs for consumer applications, with increasing returns to any new customer. Already Saxenian (1994) showed how the increasing number of firms in the Silicon Valley triggered demand and corresponding support structures. Further, Gao (2011) showed that VCs increase their rate of successful exit with diversification along the value chain (which they call the “software stack”).

Direct and indirect effects co-vary. Standards provide the most important source for both types of network externalities (Katz and Shapiro 1985). With each gained customer a firm not only increases its earnings but also raises its chances for establishing the standard to be adapted by upstream and downstream companies (indirect effects) and, thus, increases its chances of more customers (direct effects). In business to business relationships firms act as customers for other firms, due to division of labour within an industry, thus likely creating bandwagon effects (Arthur 1994; Wade 1995).

From an individual firm perspective, this calls for a strong emphasis on growth strategies (Eisenmann 2006) or for complementary product-market related diversification, which has been shown to have a positive impact on sales growth (Tanriverdi and Chi-Hyon 2008) and other performance criteria (Lee, Venkatraman, and Tanriverdi 2010).

in the software industry. Although the usual specialization benefits do exist also in these industries, the risks of relying on independent companies are even higher when developing a market within such an environment. Therefore strong support networks (Kenney et al. 2005, comparing bio-technology, semi-conductor, telecommunications industries) and cooperation among new ventures (Podoynitsyna et al. 2013) have been found especially in areas with such (indirect) network externalities.

From an investment perspective, both for corporations and for Venture Capitalists, network externalities pose an incentive to support competitors within an industry (Economides 1996, 691). The more portfolio firms a VC acquires or supports within such an industry the higher the chances of standards to be established within that industry. This should be accompanied by investments into different firms (e.g., focusing on different platforms), producing the same effect. In summary, while virtually all VCs are focussing their investments into a limited set of industries, I hypothesize in the following that they will specialize more (in terms of number of firms devoted to an industry) when deciding to move into an industry with network externalities.

Hypotheses 2: VCs moving into an industry which is characterized through network externalities will devote more portfolio firms to this industry (higher degree of specialization) compared to industries with lesser or no network effects.

It is important to note, however, that specialization in an industry does not exclude the possibility of diversification along the value chain within the industry as it is predicted through Hypotheses 1. In fact, and as a corollary, these two hypotheses together predict that diversification along the value chain will be stronger in industries with network externalities than in others.

Geographic diversification

For developing Hypotheses 1 and 2, no assumption was necessary in regard to direct cooperation of portfolio firms or about active interference of VCs into their firms. Therefore, complementarity and network externalities should work for geographically dispersed firms also. As I will show in this section, however, the above hypotheses directly link to geographical diversification. Beside general effects of geographic proximity described in the previous section, special effects are found for the exploitation of complementarity and network effects. Geographic proximity facilitates cooperation and exchange among firms. Whereas it is mostly sufficient to ensure compatibility of offerings along a supply chain with standards, norms, or formal contracts, when no indirect networks effects are to be expected, for the creation and control of complementarities and to make sure that developments along the value chain are indeed not only compatible, but mutually value enhancing, direct exchange is much more important. Although with modern communication media this is not entirely dependent on geographic proximity, decision-makers might favour to reduce geographic distance to enhance the possibility for direct exchange and cooperation and to reduce uncertainties which are especially high in regard to intended complementarities and network effects (Podoynitsyna et al. 2013) For example, push and pull factors, discussed as mechanisms for complementarity along the value chain, will work stronger among firms within a restricted region, as

compared to internationally dispersed firms. Second, network externalities not only rely on a network of technologies, but are fostered through a network of social ties. A study of new venture performance shows positive effects of strategic cooperation to exploit direct network effects (Podoyntsina et al. 2013). Such ties are strengthened by physical encounters which are more frequent when geographic distance is limited. Especially VCs which are diversifying within an industry into firms which produce complementary products will encourage cooperation between these firms to establish standards for products and services. They will also promote business between these firms, and will encourage the exchange of knowledge. For all of this, VCs will rather concentrate their investment regionally, as long as the expectation of complementarity is high and if no harmful competition is expected among firms within a VC's portfolio. In contrast, positive effects of cooperation and geographic proximity might be overshadowed by rivalry concerns of portfolio firms within an industry which compete for the same customer base. The higher propensity for cooperation under conditions of complementarity, as opposed to strategic substitution, which rather implies rivalry, has been shown, for example, in experimental games (Potters and Suetens 2009). Therefore, VCs investing into similar firms occupying the same position on the value chain (within-segment diversification), of which the main motivation would be to spread risk, will prefer to diversify these investments geographically to avoid mutual cannibalization. Therefore, with the following hypotheses I predict geographic diversification to be lowered when VCs diversify along the value chain, assuming that they expect some complementarities among investments (as predicted in Hypothesis 2), as opposed to mere within-segment diversification. The latter would rather imply concentration on a single position of the value chain, potentially inducing rivalry among firms, which is most likely the reason why previous studies found little cooperation among firms even within a concentrated region (Ben Letaifa and Rabeau 2013). To prevent the enactment of rivalry and the fear of unintended knowledge spill-overs (Ryu, McCann, and Reuer 2018), VCs investing in a single point of the value chain will rather increase their geographic diversification. Also, despite previous findings of more geographic concentration of early stage investments (Gupta and Sapienza 1992), the above theoretical considerations do not justify to predict an influence of the stage of investment on this relationship.

Also, some VCs concentrate on certain investment stages (seed, start up, etc.), with more possible active interference of VCs as angel investors (Collewaert and Khoury 2021), requiring trust (Bottazzi et al. 2016), at early stages. Such concentration might be motivated by the VCs specific competencies and by the urge to control information asymmetries, especially at early stages (Gompers 1995), therefore wanting to reduce the distance between VC and their investee firms (e.g., Lutz et al. 2013). But this does not imply an assumption for the relationship between the number of firms in which VCs invest and their geographical distribution to be affected by the stage of investment, which is indirectly supported through a meta-study on the amount of VCs' country-level activity, showing no moderating influence of the stage of investment (Dalal 2022). In summary, VCs will want to increase the propensity of direct cooperation through spatial proximity (low geographic diversification) only if rivalry concerns are low, thus if they are not concentrating to a single point of the value chain, and, in contrast, if the expectation of complementarities are high, thus if diversifying along the value chain. Further, even as Hypotheses 2 predicts that the expectation of complementarity is highest in industries

with network externalities, such as the digital industry, the following hypothesis is formulated independent of this relationship (and not conceptualizing the type of industry as a moderator), because the VC's expectation of complementarity and diversification along the value chain as a consequence might be independent from this.

Hypothesis 3: The more VCs are diversifying along an industry's value chain, that is on different stages of a single value chain, the less they will diversify geographically for such diversified portfolio firms.

Method

Sample

Data for this study were provided through ORBIS-database (Bureau vanDijk, [n.d.](#)), used in previous diversification research (e.g., Chang, Kogut, and Yang [2016](#); Lahiri and Narayanan [2013](#)). In addition to data on corporations, this data base contains information about their holdings in subsidiaries and in portfolio firms for VCs.

I test the above hypotheses by choosing two groups of VCs. First, VCs specializing in the digital (computer and software) industry and, second, those specializing in the Health/Life Sciences Industry, with the latter assumed to be not or significantly less characterized by network effects. This difference surfaces, for example, through less clustered support networks (Kenney et al. [2005](#)). According to the above description of network externalities the main selection rationale is based on the presence of standards on which products and services have to rely. While these are of utmost importance in the computer/digital industry, the life sciences industry much more is dependent on institutional forces (legal requirements, research progress, big companies, etc.) than on standards (Gilding [2008](#)). Although networks among players in the field are strong, they rather follow idiosyncratic network dynamics (homophily, follow the trend), as has been shown for biotechnology (Powell et al. [2005](#)), instead of exploiting any network externalities. Thus, network externalities are low compared to the digital industries. For categorizing VCs to one of these industries, VCs were selected if they had at least two companies within the corresponding industry (see [Table 1](#) for definition of industries and the value chains within these industries).

To capture two institutional contexts of VCs, I selected firms for these industries in US-America and in Europe, the latter representing a region with much fewer potential firms to invest in. For example, in 2012 VC in the US completed 563 deals, compared to 229 in Europe, with individual deal size almost twice as high in the US, compared to Europe (Wall Street Journal, 31 July 2013). Europe hosts few pure financial investment firms. To preserve a comparable structure of the two sub-samples I excluded the latter from the US-sample by selecting firms with industry codes (NACE Vs 2) 6612 (Security and commodity contracts brokerage), 6619 (Other activities auxiliary to financial services, except insurance), and 8211 (Combined office administrative service activities), which usually describe venture capital firms. In addition, to avoid biases through extreme cases, for both samples VCs have been excluded which had less than 10 or more than 1000 companies within their total portfolio and those which had all their portfolio companies at only one single location. This did not

affect the European sample, excluded 24 cases from the US-sample, and lead to a sample of $n_1 = 474$ VCs in US and $n_2 = 420$ VCF in Europe (total sample size, $n = 894$).

Measures

Definition of industries

Quantitative comparisons of industries mostly rely on standard industry classifications, such as SIC and NACE, which are largely equivalent. The latter (NACE Vs 2) is available in the database used here. However, pure SIC-based measures of diversification suffer from significant threats to validity (e.g., Robins and Wiersema 2003) and do not represent the distinction between diversification along the value chain and within-segment diversification proposed here. For this, the definition of Sectors by Invest Europe (European Association of Venture Capitalists (Investeurope 2007)) appeared as most appropriate, which captures the value chains (Table 1) and contains the “software stack” used by Gao (2011) for measuring within-IT-industry diversification. For both industries this includes upstream (hardware, basic chemicals), downstream manufacturing firms (software, pharmaceuticals), and services at different stages of the value chain (basic research, wholesale, retail).

This definition of industries is, first, the bases for identifying a VC as devoted to an industry if it holds at least two firms within one of its segments, which is indicated through a dummy variable (DigDum = 1 if invested in Digital Industry, 0 if invested in Life Sciences), and, second, it is the basis for the following measures of specialization and diversification. According to this definition a VC can be specialized in both industries, which means that it is represented twice in the database (taking DigDum = 1 and DigDum = 0 for the same VC) via different portfolio firms. This is the case in 15% of VCs. Because, this might only have a conservative effect on any tests of differences between industries (assuming that a single VC will less discriminate its diversification strategies between industries than different VCs), I did not exclude these cases from the analysis.

Degree of Specialization is captured through the number of VC’s portfolio firms devoted to an industry (nChain). To control for size, the total number of VC’s portfolio firms is used as a control (nSub).

Within industry diversification along the value chain

I applied a standard entropy measure to operationalize diversification along the above defined industries, as it has been used in many previous studies for corporate diversification (e.g., Robins & Wiersema, 2003) as well as for VC diversification (Matusik and Fitza, 2012).

$$DivChain = \sum p_k * \ln(1/p_k),$$

with p_k denoting the share of portfolio firms within industry segment k . However, different from previous studies, for this measure industry segments are restricted to those forming the value chain within an industry (see Table 1).

Geographic diversification

In contrast to industry diversification, for which entropy-based measures are well established, such a standard is lacking for geographic diversification. While studies focusing on

institutional differences between countries also rely on entropy based on country or regional distinctions (Cressy, Malipiero, and Munari 2014; Thams, Alvarado-Vargas, and Newburry 2016), thus abstracting from geographic distance, others use road distances (Lindgaard Christensen 2007) to measure proximity within smaller geographic region or travel time (Lutz et al. 2013) to capture the costs of visiting dispersed units. Especially if units, such as portfolio firms of a venture capital firm, are dispersed over different continents other consideration, such as time zone differences might be taken into account through more complex measures. While individual decision-makers might indeed use different calculi when deciding upon investing into portfolio firms, a single measure of geographic diversification is desired which should correlate with such individual consideration and which represents them in a most valid way. Geodesic distances between latitude-longitude positions of portfolio firms captures all of these considerations through a single measure. It highly correlates with entropy of time zones (in the data base used here: 0.36, $p < 0.01$) and with variance of country (0.69, $p < 0.01$). Also, it correlates highly with travel time in driving distances (below 500 km, as available from internet route planning programmes, $r = 0.78$, in a sample of 42 location pairs) and even higher for flying distances (as estimated by a travel agent for a sample of 40 location pairs, $r = 0.95$). Further, while some previous studies used the proximity between VCF and its portfolio firms, the measure used here is a true diversification measure in the sense that it measures only distances between portfolio firms, independent from the location of the VC's headquarter. In particular, a two-stage procedure was applied to construct a measure for the VC's geographic diversification. First, to avoid sequence effects for computation of pairwise distances VC have been sorted along their longitude position before computing the mean (thereby approximating the shortest path between all portfolio firms, for 13% of firms no location had been available in the data-base, which had been excluded from the measure). Second, from this sequence, I computed the mean subsequent pairwise geodesic distances.

$$GeoDiv = Mean_{i,j}(geodistance_{ij})$$

with geodesic distance (in km) approximated by the following formula:

$$geodistance_{ij} = \arccos[\sin \varphi_i * \sin \varphi_j + \sin \varphi_i * \sin \varphi_j [Abs[\theta_i - \theta_j]]] * 6371,$$

where φ denotes the latitude and θ the longitude (in degrees) of the two positions, and 6371 km the mean radius of the earth. Thus, a low mean (relative proximity of portfolio firms) signifies low geographic diversification. For the analysis, the logarithm of this measure is used.

Controls

Location of the VC

From a pure rationality perspective, VCs should select their portfolio firms based on potential returns independent of its own homebase. However, VCs investments are path dependent in the sense that prior investments in a region increase the probabilities for further investment (Chen, Liu, and Chen 2009). Further, even with electronic media and cheap traveling, screening and monitoring costs for investments increase with the

distance from the location of the VC. The geographical location of the VC restricts the diversification choices by building on existing networks (Guler and Guillen 2010) and by decreasing the likelihood of intensive interaction, which is necessary to build trust between the VC and its portfolio firm (Sorenson et al. 2001), similar to corporations which need to monitor their divisions (Jia and Mayer 2015). For example, in an environment like California where many start-ups compete for funding, VCs find more opportunities to select firms which fit their portfolio than in other parts of the US (Sorenson et al. 2001, 1570), or in Europe (Lutz et al. 2013). Although they could choose investments around the globe, their possibilities to actively engage in monitoring firms would be much more limited. Therefore, I expect the effects of Hypotheses 2 to 3 to be influenced by the location of the VC and the location of the VC has to be considered as a controlling variable for the above relationships. These effects will surface only to the extent that the region (state, continent) in which the VC is located provides enough opportunities to invest into different companies, which fulfil the requirements of the VCs strategy. Therefore, the Location of the VC either within the US, (East Coast versus West Coast) or within Europe (Country Dummies) controls for this effect, leading to a total of 11 regional Dummy Variables.

Total number of portfolio firms

As indicated above, the total number of VC's portfolio firms (nSub) is required to qualify the number of firms within an industry as a measure of specialization.

Within segment geographic diversification

Further, VCs might differ in regard to geographic diversification regardless of their diversification strategy along the value chain. Therefore, I used the mean within segment geographic diversification (GeoDivIntern) as an additional control:

$GeoDivIntern = \frac{1}{n_k} \sum_{k=1}^{n_k} Mean_{i,j,k}(geodistance_{ij})$ with k denoting the industry segments from Digital Industries or the Life Sciences as defined in Table 1. Again, the logarithm of this measure is used for analysis.

Syndication

There is considerable research on Venture Capital Firms' (VCF) syndication of their investments, showing effects on investments into portfolio firms (Anokhin et al. 2011; Brander, Amit, and Antweiler 2002), including cooperation activity (Ozmel, Reuer, and Gulati 2012) and geographic distribution (Sorenson et al. 2001). To control for such effects, I computed the mean (across portfolio firms) of the number of joint ownerships for each VC with other VCs in the database.

Number of shareholders of VCF

Diversification might be driven also by the diversity of interests of shareholders. Therefore, I also included the number of shareholders of VCF.

Analysis and results

The effects proposed through Hypotheses 1, 2, and 3 in the following are formulated as a system of three equations, which serves as the basis for the subsequent analysis.

$$nChain = \beta_1 * DigDum + \delta * controls + \varepsilon \quad (1)$$

$$DivChain = \beta_2 * nChain + v * controls + \rho \quad (2)$$

$$GeoDiv = \beta_3 * DivChain + \theta * controls + \psi \quad (3)$$

To estimate the parameters of these equations, I employed Two-Stage-Least Square (2SLS) regressions as implemented in STATA (Vs. 14), where independent variables in Equations (2) and (3) are instrumented by equations (1) and (2). This allows both for controlling of endogeneity as it is implied by the model, including the use of exogenous instruments. Especially for diversification, empirical tests have shown that it is affected by endogeneity due to self-selection of corporations, for example when examining impacts on firm value (Campa and Kedia 2002; Chang, Kogut, and Yang 2016). Two-Stage-Least Square, however, is not able to control for potentially correlated error terms. Therefore, an additional analysis with Three-Stage-Least-Square (3SLS) has been performed. Because 3SLS is sensitive to heteroscedasticity, potentially leading to inconsistent estimates, I performed a White test, indicating a significant deviation to homoscedasticity (Chi-Square: 60.4, $p < 0.01$). Examination of residuals suggested outliers of the dependent variables to be the reason. Therefore, I ran the estimation with a robust sample by restricting the range of the dependent variable Geographic Diversification to $geodiv > 100$ & $geodiv < 5000$, which reduced the sample by 157 observations ($n_2 = 737$). As expected, this eliminated heteroscedasticity (Chi-Square = 5.60, n.s.), while providing only slightly different estimates (non-significant in all cases) compared to the full sample. Therefore, only the full sample results will be reported in the following.

Additional sensitivity tests have been performed. One major advantage of 2SLS and 3SLS regression models is the possibility to account for potential endogeneity, possibly biasing tests of causal connections. While self-selection should be less a problem for VCs, which are diversifying by definition, and while firm value is not used as a dependent variable here, I tested first for endogeneity concerns by employing a Hausman test (Hausman 1978, 1259) on those causal connections for which this could apply. This required the identification of instrumental variables for two independent variables: for specialization ($nChain$), the total number of firms within the two-digit NACE industry ($n2S$) has been chosen, and for diversification ($DivChain$) the total diversification within this industry ($Div2S$) has been used. Entering the residuals of the reduced form, OLS-regression (Wooldridge 2010) yielded non-significant estimators for these two regressions, indicating a lack of endogeneity for neither equation. Because, 3SLS results did differ from 2SLS only in a minor and non-significant manner, I report only the latter, which require less critical assumptions.

For a further sensitivity test I, first, replaced the logarithms of geographic distance (as in Sorenson et al. 2001) with the original measure and, second, with a distance measure based on travel time to capture the different situation in US versus Europe. Because both correlated highly with the used measure (above 0.9), they did not significantly change the estimated relationships. Finally, direct relationships between network externalities and diversification measures have been tested. These direct effects all were insignificant and did not change significantly the results reported below.

Table 2. Descriptive statistics and correlations (n = 894).

		Mean	Std. Dev	1	2	3	4	5	6	7
1	Geodiv (Log)	6.42	1.47							
2	DivChain	.74	.39	.19						
3	GeoDiv	6.70	1.61	.89	.23					
	Intern (Log)									
4	nChain	10.56	12.6	-.03	.35	.11				
5	DigDum	.53	.50	.03	-.04	-.00	.05			
6	#shares	5.34	14.0	-.04	.14	.02	.17	-.02		
7	syndic	1.15	.91	.27	.19	.27	.16	.08	-.07	
8	nSub	63.93	120.0	.03	.33	.13	.73	-.01	.30	.09

Table 2 compiles descriptive statistics and correlations among variables in the final analysis. All of the hypotheses are supported through highly significant estimated relationships between the according variables (**Table 3**). Hypotheses 2 (1st Column) is supported through a positive relationship between the dummy variable indicating the industry with network externalities versus the life science industry (DigDum) with the degree of specialization (the number of firms within the industry – nChain – controlled through the total number of firms – nSub). The estimator suggests that, keeping other things equal, VCs acquire 55% more firms if they are invested in the Digital Industries than if they focus on the Life Sciences. Hypotheses 1 predicts a positive impact of specialization (nChain) on the degree of within-industry diversification along the value chain (DivChain) which is also supported (2nd column in **Table 3**). Finally, the latter (Hypotheses 3) shows a highly significant negative impact on geographic diversification.

Table 3. Results of 2SLS regressions (estimates, standard errors in parenthesis).

Variable	1.Industry specialization (nChain)	2.Within-industry diversification along value chain (DivChain)	3.Log of Geographic Diversification along the value chain (GeoDiv(Log))
Digital Industry Dummy (DigDum)	1.51 ** [0.580]	-0.065** [0.013]	
Industry specialization (nChain)		0.01** [0.001]	
Diversification along Value chain (DivChain) (Log)			-1.67** [0.364]
Controls			
# portfolio firms (nSub)	0.078** [0.002]		
Number shares	-0.043 [0.022]	0.002** [0.001]	
Amount of syndication (syndic)	1.45 ** [0.352]	0.024 [0.015]	
Within-segment geographic diversification (GeoDivIntern) (Log)			1.13** [0.314]
Country dummy	Yes	Yes	Yes
# Observations	894	894	894
R Squared	0.55	0.20	0.61
Chi-Squared	1085.2	195.1	610.3

**p <0.01.

Supplementary analysis and robustness test

Although not of prime interest here and used as a control, the mentioned effects of the VCs location (US vs. Europe) deserves special attention, because they suggest, that the proposed relationships especially surfaces within a region with high munificence in regard to investment opportunities, whereas it shows only weakly in Europe and other regions, while still holding theoretically. Therefore, I tested the model in addition separately for the European sub-sample. Estimates for proposed relationships remained in the same direction and significant, only showing a higher significance level ($p < 0.001$) for DigDum (influence of network externalities) and a slightly weaker, but still significant impact ($p < 0.05$) of Diversification along the value chain (DivChain) on Geographic Diversification (GeoDiv). Additional sensitivity tests for the model included the examination of direct relationships between variables (Industry dummy and Degree of Diversification and Degree of Geographic Diversification) in addition to the proposed relationships. Estimates for these direct relationships were neither significant nor improved the fit of the overall model.

Finally, I checked whether the proposed and found *negative* relationship between diversification along the value chain and geographic diversification does also appear when the former is replaced through within-segment diversification (regardless of the value chain), which would threaten the significance of Hypothesis 3 and the present result. As expected, however, the according model estimation shows a significant *positive* relationship (0.33 [0.06], $p < 0.01$) between within-segment diversification and corresponding geographic diversification, which clearly contrasts the negative relationship between the corresponding measures along the value chain.

Discussion

Contributions

The present study is the first to link industry diversification choices to geographic diversification, based on the theoretical assumption that VCs consider complementarity of its investments for selection of investments and for their geographical location to exploit supermodular return functions. For this, the paper focussed on diversification along an industry's value chain as opposed to pure within-segment diversification. Thus, it contributes to the fields of strategy and geographic agglomeration.

First, the paper provides evidence for investment patterns, which are based beyond mere individual firm considerations. In a highly significant number of cases VCs not only aim at nurturing individual firms, but groups of seemingly competing firms. As the paper shows, this is achieved by diversifying along the value chain, rather than through within-segment diversification. By showing the explanatory power of this distinction the paper expands existing research of within-industry diversification (Hashai 2015; Li and Greenwood 2004; Tanriverdi and Chi-Hyon 2008). The paper shows that VCs may "build" firms by simply "picking" (Baum and Silverman 2004) those which are complementary.

Second, network externalities so far have been mainly studied in single industries. By comparing two industries, this study relates network effects to VCs specialization decision. Obviously, VCs are aware of network externalities when moving into such industries. As

a consequence, and as it has been shown here, they will devote more investments to such industries and, subsequently, shape their diversification strategy accordingly. If the necessary number of players is missing at the beginning, VCs play an active role by investing into new start-ups (“picking”), nurturing existing firms, and by promoting interaction among these firms (“building”).

Third, the latter leads to the investigated relationships with geographic diversification. While geographic agglomeration research is based on sound theoretical arguments, it failed to relate geographic diversification decisions to other strategic considerations on the firm level so far. The present paper provides strong evidence that location decisions are motivated by the aim to create supermodular return functions and, thus, are combined with diversification along the value chain of an industry. VCs are hardly competition seeking for their portfolio firms, as it might be the case under pure demand side considerations. Rather, they invest into competing firms only if they are located in different areas of the globe. On the opposite end, they will concentrate their investments geographically if firms produce mutually reinforcing products or services. Still, regional clusters, as a main outcome of geographic agglomeration, might direct the search for investment if already existing. Most important, however, the paper calls for a more nuanced view on supply side effects of regional agglomeration. It suggests that, for example, the existence of specialized human resources and capabilities in clusters is hardly sufficient to justify geographic concentration for directly competing firms. Regional concentration involves both risks for VCs, for example, by potentially restricting the search for talent, and opportunities, for example, by enabling the use of local networks. The paper, however, provides evidence, that all such considerations will apply especially if firms’ products or services do not directly compete for the same customer base, but are mutually reinforcing.

Further, the study differs from most previous empirical research not only by comparing two industries but by including samples from two different institutional contexts. This is based, on the one hand, on the assumption that the theoretically derived hypotheses of the model apply to both contexts, but that, on the other hand, they might surface in different ways empirically, because of exogenous conditions which more or less facilitate the implementation of strategies. The sub-sample analysis showed that the relationships hold in the same way for European VCs as for US-based VCs, even as the former might find it more difficult than the latter to implement their strategies by deliberately restricting their investments to their region.

Building theoretically on the notion of complementarity and supermodularity (Milgrom and Roberts 1995) and by showing how this may predict and explain investment patterns of VCs, the paper suggests also to include such considerations and distinctions of diversification strategies, not only for VCs but also for corporations. Supermodularity may explain diversification patterns beyond the value chains used here and beyond the well-researched related-unrelated continuum. While complementarity has been acknowledged for predicting diversification even from a resource-based perspective (Wan et al. 2011, 1350), it may explain the performance of unrelated diversified corporations as an exogenous variable without or in addition to considering complementarity of resources.

Managerial implications

While this paper did not examine effects on performance criteria, the theoretical derivations suggest that VCs should analyse their industries chosen as specialization areas in regard to complementarity and supermodularity, before investing into individual firms. This should supplement traditional advice for VCs which base their decision mainly on individual firm characteristics, such as features of the product and assessed management competence (Miloud, Aspelund, and Cabrol 2012). While this remains important, supermodularity might raise or lower the value of an individual firm when valuing it together with other investments of the VC itself or even with those of other VCs in the region. Thus, the VC would act beyond its identity as a firm, but together with its portfolio as a quasi-firm (Luke, Begun, and Pointer 1989). Still the VC's investment pattern is hardly stable over time, because of systematic exits at pre-defined development stages of portfolio firms. Also, while Luke and his colleagues (Luke, Begun, and Pointer 1989) exemplified the quasi-firm for the health care industry, largely by referring to institutional pressures, such might be even more called for in industries characterized through network externalities where institutional demands are comparatively low, but where the effects of supermodular value functions are stronger.

Further, while this paper focussed a clearly exogenous type of supermodularity (the value chain) I maintained already above that this is hardly the only possible form and, thus, VCs should treat supermodularity not just as exogenous, but might be included as endogenous to their decisions. Through their investment patterns VCs (and corporations as well) potentially only create those conditions for complementarities which before did not exist. Consider the recent example of digital currencies. Like traditional currencies, they are all-purpose means. No special complementarities exist. However, by designing financial applications (especially based on the blockchain technology) which are dependent on the existence of digital currencies, these two product categories become complementary – a supermodular value function has been created which did not exist before. As the paper in addition suggests, VCs should benefit from a nuanced approach to geographical diversification beyond the path dependences and beyond the focus on industrial clusters. While the latter might be a complexity reducing vehicle to focus complementary firm investments, within segment diversification should maximize the resource pool by diversifying around the globe, even overcoming existing restrictions through the VC's own homebase.

Limitations and future research

The literature provides groupings for VCs which might be relevant in regard to their diversification decisions, including variables such as age, size, experience, innovativeness, and which have not been controlled in this study. This is the case especially for differences between active versus passive VCs, a distinction which attracted already some empirical studies (Bottazzi, Da Rin, and Hellmann 2008; Inderst and Mueller 2009), for example relating it to the stage of investment (Berglund 2011), but not to other strategic choices. It is highly probable that the VC's geographic location has an increased influence on its geographic diversification if it is an active investor, because their ability to monitor, control, and advice their portfolio firms decreases across large distances. Future studies

should incorporate such distinctions, potentially increasing the explained variance for geographic diversification. Also, the interpretation of the VC together with its portfolio firms as a quasi-firm, put forward above, could be contested by examining the strategic influence of VCs on their portfolio firms.

Depending on their share of total investments for newly founded enterprises within a region, VCs geographic concentration should be examined in longitudinal studies to shed light on their influence for the creation and growth of regional clusters. Again, significant differences between the US and other regions like Europe and Asia should be expected, with different institutional environments, government influence and cultural idiosyncrasies, beside the usually considered supply side effects.

As has been mentioned already above, to derive managerial implications diversification strategies should be related to VCs' returns, like it has been done in few previous studies (Cressy, Malipiero, and Munari 2014; Gao 2011) and which might include innovation related criteria in addition to pure financial performance measures (Lahiri and Narayanan 2013). Especially needed would be the incorporation of risk measures which are associated with diversification strategies of VCs, building on the assumption that diversification along the value chain also imposes less risk than within-segment diversification.

Although the results surely are limited to VCs' diversification, hypotheses and subsequently managerial implications for corporations and corporate venture capital (Yang, Narayanan, and De Carolis 2014) are possible and deserve future research. When vertically integrated, especially in an industry with network externalities, cooperative structures should be superior to a pure competitive divisionalized structure. Finally, managerial implications could be drawn even for the level of individual portfolio firms, which are essentially bearing the risks of VCs' diversification strategies. High risks of such strategies imply high likelihoods of failing for individual firms. Thus, if future studies prove that diversification along the value chain impose relatively low risks, new firms scanning the market for venture capital firms may rather prefer VCs diversifying along the value chain over other types of venture capital firms.

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